

State of Developer Ecosystem 2024: Language Trends, Satisfaction, and

Key insights from JetBrains' survey of 23,262 developers on language usage (2017–2024), satisfaction, adoption intent, and domain specialization

What you'll get in 6 slides

Survey size

How language usage is shifting over 8 years

23,262 developers

Which languages developers love most (satisfaction)

How languages map to application domains (specialization)

Practical takeaways for 2025 technology strategy

Time window

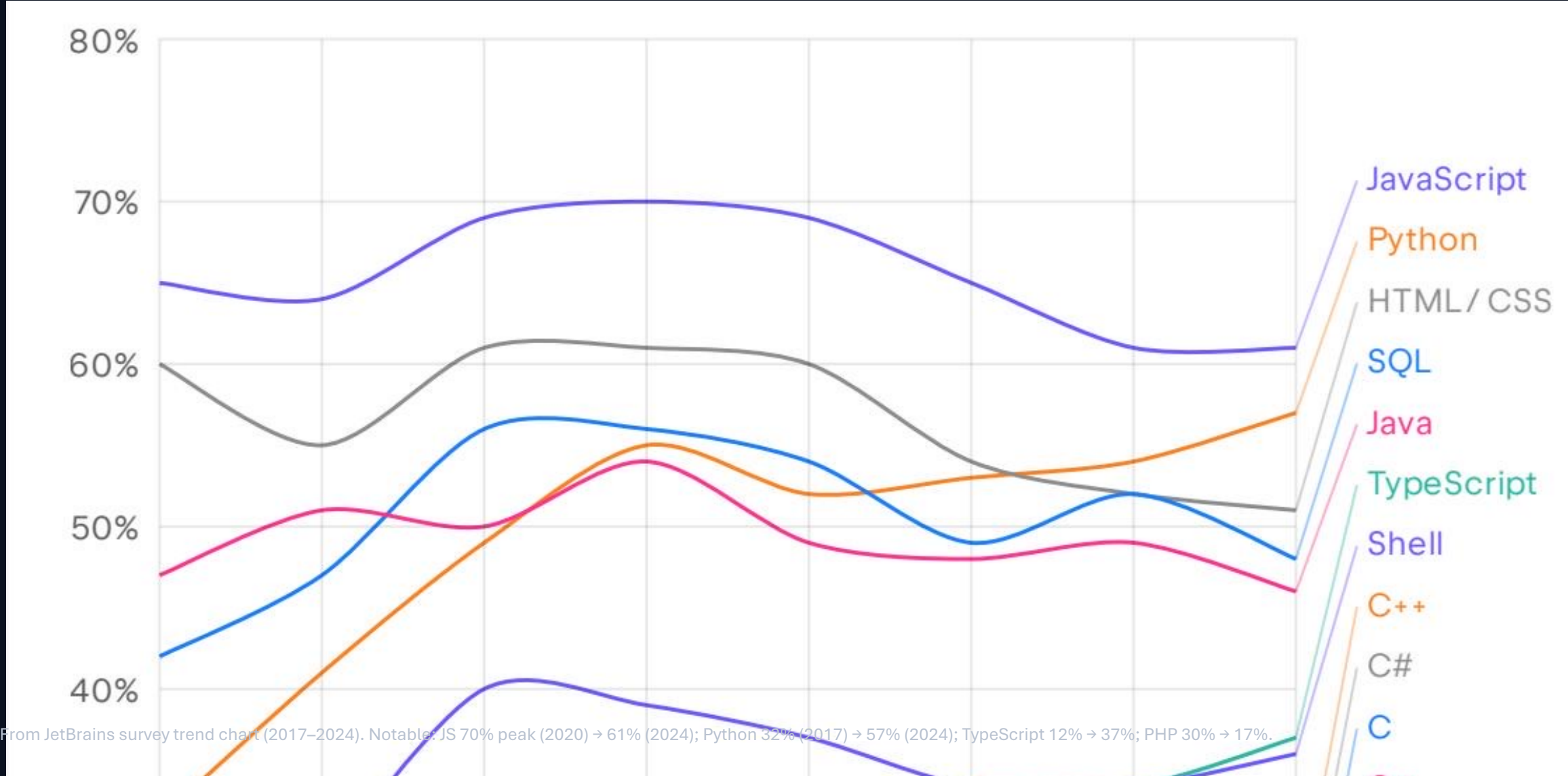
2017–2024 usage

Audience

Tech strategy & hiring

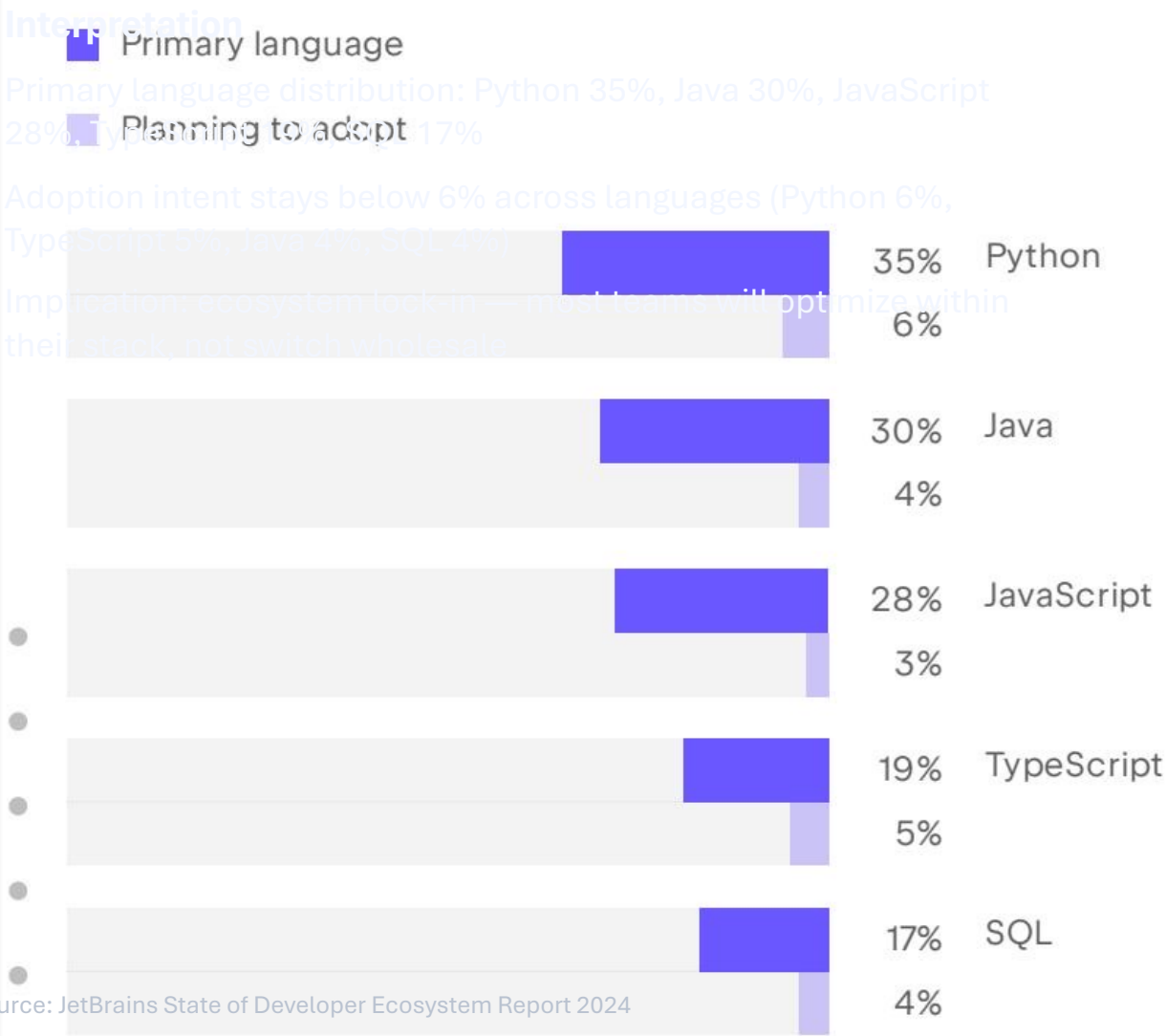
Python Closes the Gap: 8 Years of Language Usage Shifts

Usage share among developers, 2017–2024



Python Is Now the #1 Primary Language at 35%

Primary language vs. plan to adopt



TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198

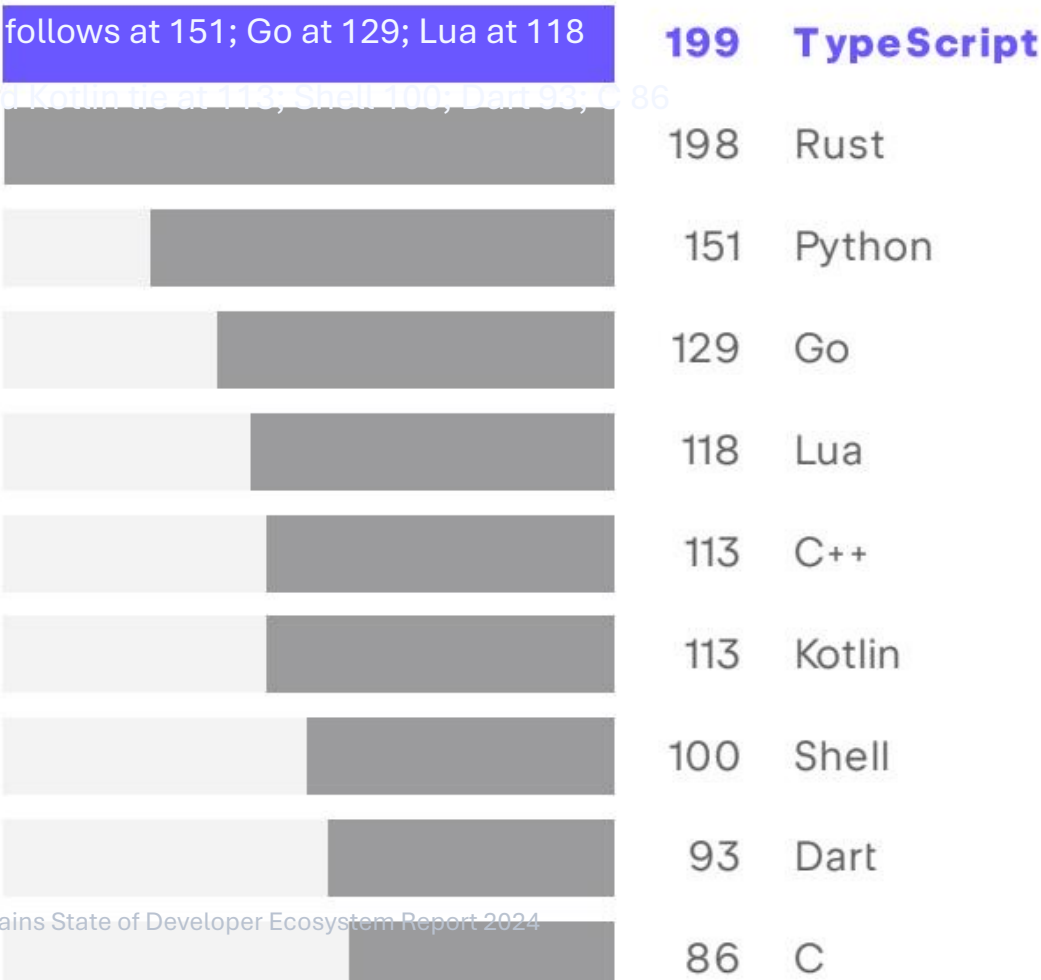
Satisfaction score ranking (top 10)

What stands out

TypeScript (199) and Rust (198) lead — modern DX + safety

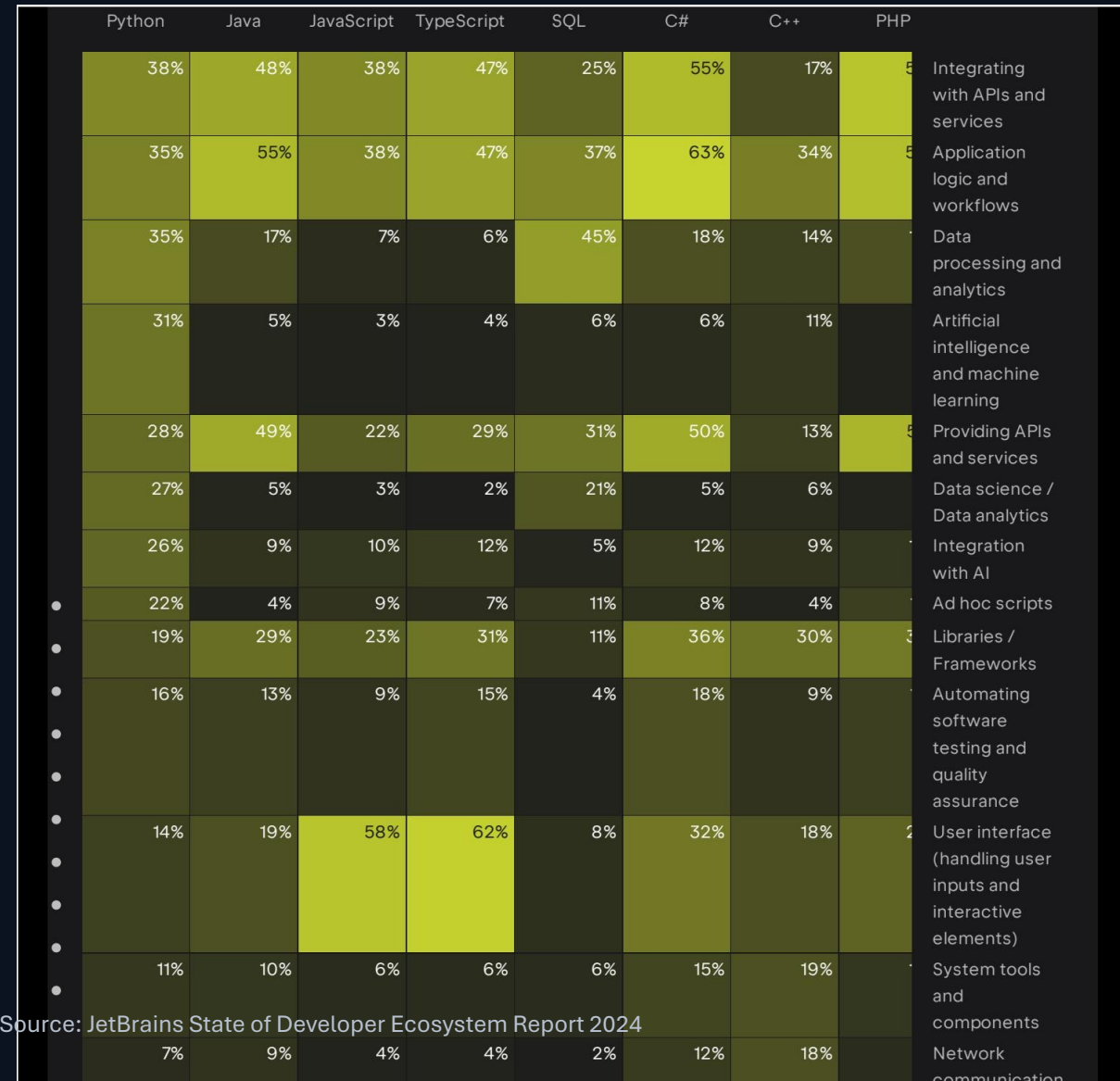
Python follows at 151; Go at 129; Lua at 118

C++ and Kotlin tie at 113; Shell 100; Dart 93; C 86



Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Top language by application area



Source: JetBrains State of Developer Ecosystem Report 2024

Domain	Top language	Share
AI / Machine Learning	Python	31%
Data Processing & Analytics	Python	17%
User Interface	TypeScript	62%
APIs & Services	Java	49%
Application Logic & Workflows	Java	55%
System Tools & Components	C++	27%
Integrating with APIs & Services	Python	38%

What This Means for Your Technology Strategy in 2025

Actions for platform, product, and hiring

Invest in Python for AI/ML

Python leads AI/ML (31%) and data science (28%); it is also #1 primary language (35%).

Standardize TypeScript for frontend

TypeScript dominates UI (62%) and tops satisfaction (199).

Plan for systems growth

Rust satisfaction (198) and steady usage growth; Go stabilized around 18% usage after rapid growth.

Be cautious on declining pools

PHP fell from 30% usage (2017) to 17% (2024); Ruby fell from 10% to 4%.

Retention ties to DX

Top satisfaction clusters around type safety + modern tooling; factor into developer experience decisions.